Functional Programming – Lambda Calculus Exercises

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Exercises

1. Show that $(\lambda\,\mathrm{x}.\,\lambda\,\mathrm{y}.\,\mathrm{x})\;(\lambda\,\mathrm{x}.\,\mathrm{x})(\lambda\,\mathrm{z}.\,\mathrm{z}\;\mathrm{z})(\lambda\,\mathrm{z}.\,\mathrm{z}\;\mathrm{z})$ reduces to $(\lambda\,\mathrm{z}.\,\mathrm{z}\;\mathrm{z})$ by lazy evaluation.
2. With Church Booleans prove that AND defined by $\mathrm{AND} \equiv (\lambda\,\mathrm{a}.\,\lambda\,\mathrm{b}.\,\mathrm{a}\;\mathrm{b}\;\mathrm{FALSE})$ has the expected behaviour.
3. Show that under both strict and lazy evaluation $\mathrm{I}\;\mathrm{M} \neq \mathrm{M}\;\mathrm{I}$ (where I is the identify function and M is the mockingbird function)

Solution 1

- $\underline{(\lambda x.\, \lambda y.\, x)(\lambda x.\, x)}(\lambda z.\, zz)(\lambda z.\, zz)$

  $\xrightarrow{\beta}$ $\underline{(\lambda\, \mathrm{y}.\, \lambda\, \mathrm{x}.\, \mathrm{x})\ (\lambda\, \mathrm{z}.\, \mathrm{z}\ \mathrm{z})}(\lambda\, \mathrm{z}.\, \mathrm{z}\ \mathrm{z})$

  $\xrightarrow{\beta}$ $\underline{(\lambda\, \mathrm{x}.\, \mathrm{x})\ (\lambda\, \mathrm{z}.\, \mathrm{z}\ \mathrm{z})}$

  # the argument $(\lambda\, \mathrm{z}.\, \mathrm{z}\ \mathrm{z})$ is dropped because y is not part of the function body

  $\xrightarrow{\beta}$ $(\lambda\, \mathrm{z}.\, \mathrm{z}\ \mathrm{z})$

  # The identity function just returns its argument unchanged

Solution 2

- Remember
  - $\text{TRUE} \equiv (\lambda\, \text{x}.\, \lambda\, \text{y}.\, \text{x})$ (takes two inputs, returns the first)
  - $\text{FALSE} \equiv (\lambda\, \text{x}.\, \lambda\, \text{y}.\, \text{y})$ (takes two inputs, returns the second)
- There are 4 cases AND TRUE TRUE, AND TRUE FALSE, AND FALSE TRUE, AND FALSE FALSE
- $\text{AND TRUE TRUE} \equiv \underline{(\lambda\, \text{a}.\, \lambda\, \text{b}.\, \text{a b FALSE})\ \text{TRUE}}\ \text{TRUE}$

$$\xrightarrow{\beta} \underline{(\lambda\, \text{b}.\, \text{TRUE b FALSE})\ \text{TRUE}}$$

$$\xrightarrow{\beta} \text{TRUE TRUE FALSE}$$

$$\equiv \underline{(\lambda\, \text{x}.\, \lambda\, \text{y}.\, \text{x})\ \text{TRUE}}\ \text{FALSE}$$

$$\xrightarrow{\beta} \underline{(\lambda\, \text{y}.\, \text{TRUE})\ \text{FALSE}}$$

$$\xrightarrow{\beta} \text{TRUE}$$

- And similar for the other cases

Solution 3

- Lazy eval:

$$IM \equiv (\lambda x.\, x)(\lambda z.\, zz)$$
$$\xrightarrow{\beta} (\lambda\, \mathrm{z}.\, \mathrm{z}\ \mathrm{z})$$
$$\mathrm{MI} \equiv (\lambda z.\, zz)(\lambda x.\, x)$$
$$\xrightarrow{\beta} (\lambda\, \mathrm{x}.\, \mathrm{x})\ (\lambda x.\, x)$$
$$\xrightarrow{\beta} (\lambda\, \mathrm{x}.\, \mathrm{x})$$

- Strict eval:
  - Behaves exactly the same because if there is only a single application with no nested functions then there is only one possible order of β-reduction.
  - Anyway, in general, if both orders terminate they will produce the same final result.